Duarte Rocha

+31(0)643538333 | duartefgrocha@gmail.com

Utrecht, The Netherlands

duarterocha17 | duarterocha | Duarte Rocha

Personal description and objective

PhD Researcher in Fluid Dynamics with a passion for solving physics problems from fundamental principles. I thrive on tackling challenging problems by collaborating with talented colleagues. I am detail-oriented and scientifically curious, I value in-depth discussions for progress and innovation. Beyond research, I maintain an active lifestyle through CrossFit, biking, and team sports, and enjoy traveling to foster creativity and work-life balance.
My research focuses on evaporation-driven flows and interfacial phenomena in multicomponent droplets at the microscale, investigated through Computational Fluid Dynamics (CFD), using Finite Element Method (FEM) software. I am looking to translate fundamental fluid dynamics into robust numerical tools for industrial R&D.
Expected PhD completion: September 2026.

Experience

- **Researcher PhD student** — *Sep 2022 - Present*
  Physics of Fluids Department | *University of Twente* — Enschede, The Netherlands
  - Supervisors: Christian Diddens and Detlef Lohse. Part of the Industrial Partnership Programme *"Fundamental Fluid Dynamics Challenges in Inkjet Printing"*, in collaboration with *"Canon Production Printing"*.
  - Co-developed PYOOMPH, an open-source object-oriented multi-physics FEM framework in Python, using Git for version-controlled collaborative development.
  - Published 7 peer-reviewed papers in high-impact journals (see publications section), presented at 7 international conferences (e.g., APS–DFD), and attended 4 specialist fluid dynamics and high-performance computing courses.
  - Developed high-fidelity simulations of microscale droplet physics, capturing interfacial and evaporation phenomena with precision sufficient to resolve Marangoni instabilities and symmetry-breaking transitions.
- **Master's Thesis Research Fellow** — *Jan 2022 – July 2022*
  Transport Phenomena Research Center | *University of Porto* — Porto, Portugal
  - Supervisors: Francisco Galindo-Rosales and José Daniel Araújo. Designed and ran CFD simulations using ANSYS Fluent to optimise nozzle geometry for electronic printing applications, identifying design parameters that improve flow stability and contributing to enhanced printing precision.
- **Mechanical Engineer Intern** — *Jun 2019 – Aug 2019*
  IKEA Industry Portugal — Paços de Ferreira, Portugal
  - Analysed and optimised the mechanical design of a component detection system for packaging lines, improving robustness and reliability of detection by 90% under industrial operating conditions.

Education

- **Master's Degree in Mechanical Engineering** — *Sep 2017 - Jul 2022*
  Faculty of Engineering of the University of Porto — Porto, Portugal
  - Specialisation in Fluids and Energy. Final Grade: 85%.
- **Mobility Programme** — *Sep 2020 - Jul 2021*
  Faculty of Civil and Transport Engineering of Politechnika Poznańska — Poznań, Poland
  - Erasmus studies. Final Grade: 95%.

Skills

- **Technical:** PYOOMPH, Python, C++, MATLAB, Ansys Fluent (CFD), AutoCAD, SolidWorks, Git.
- **Methods & Fields:** FEM, CFD, Numerical Analysis, Inkjet Printing, Microfluidics, Heat Transfer.
- **Languages:** Portuguese (Native - C2), English (Full Professional Proficiency - C1), Spanish (Limited Working Proficiency - B1), Dutch (Basic - A2).

Peer-reviewed Publications

7 peer-reviewed publications (3 as main author), spanning *Journal of Fluid Mechanics*, *Journal of Computational Physics*, *Advanced Science*, and *Soft Matter*.

[1] Diddens, C. & Rocha, D. (2024). Bifurcation tracking on moving meshes and with consideration of azimuthal symmetry breaking instabilities. *Journal of Computational Physics*, 518, 113306. doi:10.1016/j.jcp.2024.113306

[2] Meijer, Jochem G., et al. (2024). Enhanced bubble growth near an advancing solidification front. *Journal of Fluid Mechanics*, 996, A22. doi:10.1017/jfm.2024.777

[3] Rocha, D., et al. (2025). Evaporating sessile droplets: solutal Marangoni effects overwhelm thermal Marangoni flow. *Journal of Fluid Mechanics*, 1013, A39. doi:10.1017/jfm.2025.10208

[4] Rocha, D., et al. (2025). Marangoni flow driven hysteresis and azimuthal symmetry breaking in evaporating binary droplets. *Journal of Fluid Mechanics*, 1018, A11. doi:10.1017/jfm.2025.10519

[5] Gore, S., et al. (2025). Symmetry Breaking in Chemical Systems: Engineering Complexity Through Self-Organization and Marangoni Flows. *Advanced Science*, e15672. doi:10.1002/advs.202515672

[6] Bisswanger, Steffen, et al. (2026). Upstream motion of oil droplets in co-axial Ouzo flow due to Marangoni forces. *Soft Matter*, 22, 567–577. doi:10.1039/d5sm00848d

[7] Dekker, P. J., et al. (2026). Internal flow and concentration in neighbouring evaporating binary droplets and rivulets. *Journal of Fluid Mechanics*, 1032, A33. doi:10.1017/jfm.2026.11339